Manas Choure

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Education

- **University of California Los Angeles** — Los Angeles, CA
  Doctor of Philosophy in Electrical Engineering ([redacted] — *June 2026 - May 2030 (projected)*
- **University of Pennsylvania** — Philadelphia, PA
  Bachelor of Arts in Economics and Physics — *August 2022 - May 2026*
  Master of Science in Physics; GPA: 3.90 — *August 2024 - May 2026*
  Activities: Cricket Club at Penn, Science Olympiad at Penn, Society of Physics Students
- **Coursework:** Quantum Mechanics and Computing, Intro to Data Structures with **Java**, Physics Computing in **Python**, Electromagnetism, Data Analysis in Economics with **R**, Macro-econometrics, Nonlinear Optics

Experience

- **Atomic Clocks and Device Physics Intern** — El Segundo, CA
  The Aerospace Corporation — *May 2025 - August 2025*
  - **Full-Time Internship over summer, 50 Hours/week**
  - Updated serial command interface to streamline ground-to-flight integration for Optical Frequency Combs; verified autonomous startup and locking procedure with documented procedures and lab diagnostics.
  - Built and aligned free-space optics for rubidium lamp buffer gas experiments; set up experiment involving reduction in PM-AM conversion using Vertical Cavity Surface Emitting Lasers (VCSELs).
  - Learned physics behind laser lock-in systems and applied it to generate a clock signal. Started experiment from scratch and learned how to walk a laser beam to target with ample degrees of freedom.
  - Built C-based image capture for Hamamatsu detector to successfully identify 50 keV x-rays from Inverse Compton Scattering source.
  - Designed science posters for diverse audiences; gave adaptive lab tours to high school students, customers, and professionals alike.
  - Co-led career panel for Zero Robotics, coordinating industry speakers and developing questions to engage middle school students.
- **Data Research Analyst** — Fort Meade, MD
  Defense Counterintelligence Security Agency (DCSA) — *June 2024 - April 2025*
  - **Full-Time Internship over summer, 50 Hours/week; Part-Time Internship during school year, 10 Hours/week**
  - Enhanced data accuracy and streamlined data-sharing processes by 50% across mission-critical systems.
  - Identified and mitigated data errors by implementing machine learning models, improving adjudicative decision-making by 70% in background investigations.
  - Provided subject matter expertise (SME) advice in Posit systems and processes in Python, R, C++, Jupyter Notebook, and Java.
- **Teaching Assistant** — Philadelphia, PA
  Mathematics Department at University of Pennsylvania — *August 2023 - Present*
  - **Part-Time Internship over school year, 10-15 Hours/week**
  - **TA Quality:** Fall Semester - 3.89/4.00; Spring Semester - 3.91/4.00, highest ever rating in MATH 1070, received "Good Teaching Award" four times in a row for excellence in teaching in the Mathematics Department.
  - Leading recitations for 35+ students twice a week, holding office hours for 180+ students overall in the course twice a week.
  - Revising and correcting the textbook content, ensuring accuracy and clarity; positively impacting student comprehension and resulting in a 10% increase in exam scores compared to other TA sections.
- **Candidate** — Quantico, VA
  Marine Corps Officer Candidate School — *May 2023 - July 2023*
  - **Full-time Program during summer**
  - Completed 6 week session in process to become a commissioned officer of the Marine Corps.
  - Led a 40 person platoon-level group through challenging circumstances, leveraging strategic decision-making and effective communication to navigate high-pressure situations and accomplish mission objectives (e.g. combat operations, training exercises).

Publications and Presentations

- Kelley, M. J., Leifer, S., Bry, C., Thai, N., **Choure, M.** "CombSat: Frequency comb optical time transfer in a thermally unstable environment for space applications" (*in progress*)
- Kettering, H., Monahan, D., Householder, A., **Choure, M.** (2026, January). Effects of Radiation Testing on MEMS Oscillators [Presentation]. Precise Time and Time Interval Systems and Applications Meeting, Anaheim, California.
- **Choure, M.** (2025, July). Thor and Loki: A Study in Frequency Combs [Presentation]. Aerospace Technical Briefing, El Segundo, California.

Leadership

- **President and Tournament Director** — Philadelphia, PA
  Science Olympiad at the University of Pennsylvania — *August 2022 - Present*
  - **Part-Time Volunteer during school year (10-15 Hours/week)**
  - Tournament Director of the 2025 Science Olympiad at the University of Pennsylvania Invitational, managing 1200+ high school students and volunteers.
  - Received distinction of running the largest tournament in the entire country and recognition from the national chapter.
  - Handling over $25,000 per year to run events, book spaces, and organize food for volunteers and students alike.
  - Writing tests and supervising events relating to physics and math concepts at various top colleges and state-level invitationals (MIT, Yale, Penn, Georgia Tech, Princeton), authoring 24 tests to date.
- **President** — Philadelphia, PA
  Cricket Club at the University of Pennsylvania — *April 2023 - Present*
  - **Semi-Professional during school year (10 Hours/week)**
  - Conducts administrative, financial, and budgeting duties, handling $5000+.
  - Play on nationals-qualifying team, statistics of 14 wickets in 17 matches with lifetime economy rate of 5.36.
  - Arranges and leads biweekly practices with 20+ players per practice.
  - Facilitate grassroots cricket initiatives at Germantown Cricket Club and Philadelphia Cricket Club.
- **Placement Leader** — Philadelphia, PA
  Urban Initiative at the University of Pennsylvania — *October 2022 - Present*
  - **Part-Time Volunteer during school year (2 Hours/week)**
  - Set-up and direct in-person and Zoom meetings with 3 local Philadelphia high schools for Science Olympiad tutoring.
  - Teach science concepts such as optics, magnitude estimation, code-breaking, astronomy, and physics to groups of 20 students weekly.

Honors and Awards

- NSF Quantum Fellow: Awarded [redacted] for 2 years of Graduate Study - Fall 2026 to Spring 2028
- Robert J. Schweich Scholar (University of Pennsylvania): Awarded [redacted] - Fall 2022 to Spring 2026
- DCSA Student Experience **Top Presentation** (1st out of 75) - Summer 2024
- **Good Teaching Award** from the Mathematics Department at the University of Pennsylvania - Fall 2023 and 2024, Spring 2024 and 2025.
- Georgia Governor's Honors Program Finalist in Mathematics and Latin- Summer 2022
- National Merit **Finalist** - Summer 2022

Technical and Soft Skills

- **Proficient in:** Java, Python, R, AWS, Posit, LaTeX, MATLAB, C++, MS Office, Linux, Assembly.
- **Languages:** English, Hindi, Latin, Urdu, Punjabi.